

Internships, Co-ops, and How to Actually Get One

Internships and co-ops are the highest-octane version of this whole plan: paid, credentialed, engineering-adjacent work that converts directly into full-time offers. This guide covers what they are, the one gate that controls access to most of them, and the paths that are open **right now** to a high school graduate.

The vocabulary

Term	What it means	Typical length	Paid?
Internship	A temporary position, usually summer, doing junior-level real work	10–12 weeks	Engineering ones almost always pay (often \$18–30+/hr)
Co-op	A formal partnership between a college and employers: you alternate full-time school semesters and full-time paid work semesters, or work part-time year-round	3–12 months per rotation, often multiple rotations	Yes, and often better than internships
Pre-apprenticeship	A short program (weeks to a few months) that preps you for and feeds into a registered apprenticeship	4–16 weeks	Sometimes paid, usually free to attend
Youth/registered apprenticeship	Paid job + structured training from day one (see the education-pathways guide)	1–5 years	Yes, from day one

The gate nobody tells you about

Most engineering internships and co-ops require you to be a currently enrolled student. Not a graduate, not “planning to enroll” — enrolled. It’s how companies filter, it’s how their intern programs are legally structured, and it’s why a motivated 18-year-old applying to “Engineering Intern — Summer 2027” postings with no enrollment gets auto-rejected while a mediocre sophomore gets an interview.

The move this implies: enrolling at community college — even one or two evening classes while working — flips the switch. The day you have a student ID and an expected graduation date:

- You can put “Pre-engineering student, [College], expected [year]” on your resume
- You qualify for **Handshake** (the college-recruiting platform nearly every school uses — ask the college’s career center for access; local employers post there specifically to find students like you, with far less competition than Indeed)
- The college’s **career center and co-op office** start working for you for free — ask directly: “which local companies take co-op students from here?” They keep a list.
- Engineering internship postings stop auto-rejecting you

This is worth saying plainly: **one \$200 evening class can unlock a \$25/hr summer internship**. Enrollment is not just degree progress — it's a key.

What's open BEFORE enrollment (right now, with a HS diploma)

1. **Registered apprenticeships and pre-apprenticeships** — no enrollment needed; apprenticeship.gov by ZIP, IBEW locals, and community colleges (ask who sponsors “mechatronics” or “advanced manufacturing” students).
2. **Manufacturing summer/seasonal hires at local firms** — small manufacturers staff up seasonally; a summer as a production helper at an engineering firm IS an internship in everything but the title. Week 3's company census is the target list.
3. **The self-made internship** (see the pitch below) — small companies don't have intern *programs*, which means they also don't have intern *requirements*. The owner can just say yes.
4. **State workforce programs** — many states run paid youth/young-adult work programs (search “[your state] youth apprenticeship” and “[your state] workforce development young adult”); your local American Job Center (careeronestop.org finds it) can enroll you in some of these for free.
5. **A regular technician job** — never forget that the Week 2/3 technician path *is* the working person's internship: paid, full-time, and it doesn't end in August.

The self-made internship pitch

Small machine shops, controls integrators, and repair outfits (5-50 people) rarely post internships but often say yes to this, in person or by short email, addressed to the owner or shop manager:

“I'm starting an engineering degree at [college] in [month]. Before that, I want real shop experience more than I want a mall job. Would you take me on for the summer — sweeping, deburring, organizing stock, whatever's useful — so I can learn how a real shop runs? I'm reliable and I'll show up every day.”

Why it works: it asks for the bottom rung explicitly, signals reliability (their actual biggest hiring problem), and has an end date so it's low-risk for them. Ten asks will get a yes. That yes goes on the resume as *“Shop Assistant, [Company] — supported machinists in a production environment.”*

Once enrolled: playing the internship game properly

- **Timeline is everything.** Summer engineering internships at big companies open applications the **August-October before** and are largely filled by February. Mark the calendar; “I'll apply in spring” means “next year.”
- **Where:** Handshake first (least competition, local employers), then company careers pages directly (defense contractors, utilities, and manufacturers from the Week 3 census all run intern programs), then Indeed/LinkedIn with “engineering intern” + your area.

- **Community college students CAN get these.** Big employers (especially defense and semiconductors) explicitly recruit community college and engineering-tech students for technician internships. The GPA + a technician work history from this program beats a university sophomore's empty resume more often than you'd think.
- **Co-op vs. internship when transferring:** when Week 5's college research happens, ask each program: "do you have a co-op program, and which employers take your students?" A school with a real co-op pipeline is worth ranking higher — co-op students routinely graduate with 12+ months of paid engineering experience and a standing job offer.
- **The referral shortcut still rules.** A Week 7 contact saying "we take interns, send me your resume" skips the entire pile. Every informational interview should end with: *"Does your company take interns or co-op students? Who runs that?"*

How this fits the 8-week program

- **Week 3:** when building the openings list, also flag any company that mentions interns, co-ops, or apprentices anywhere on its careers page.
- **Week 5:** ask every college about co-op programs and Handshake access; ask the community college which employers sponsor students.
- **Week 7:** add the intern question to every interview (wording above).
- **Week 8:** the five-year plan should name the first internship/co-op window you'll be eligible for and the date applications open for it.